Renewable Energy 219 (2023) 119470

Contents lists available at ScienceDirect

Renewable Energy

journal homepage: www.elsevier.com/locate/renene

Machine learning based dynamic super twisting sliding mode controller for increase speed and accuracy of MPPT using real-time data under PSCs

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ARTICLE INFO

Keywords:
Maximum power point tracking (MPPT)
Partial shading conditions (PSC)
Photovoltaic (PV) systems
Machine learning
Super twisting sliding mode controller (STSMC)
Gaussian process regression (GPR)

ABSTRACT

Photovoltaic (PV) systems are created according to the series, parallel or series-parallel connection type of PV panels. Solar panels that form the PV system can have different irradiance values at different times of the day. This condition, which is called partial shading, causes the current and voltage values produced by the panels to fluctuate. Maximum power point tracking (MPPT) algorithms are used to ensure that PV systems operate at maximum power under all environmental conditions. MPPT algorithms are used to optimize the duty cycle of DC-DC converters in order to transfer maximum power to the load. In this study, a two-stage structure is used for MPPT. In the first stage, a reference voltage value is generated using the gaussian process regression (GPR) machine learning algorithm. In the second stage, the duty cycle of the PWM signal required for MPPT is optimized using the super twisting sliding mode controller (STSMC). The performance of the proposed method for three different scenarios is compared with the cuckoo search algorithm (CSA) from metaheuristic optimization algorithms and the perturb and observe (P&O) algorithm from classical optimization algorithms using real-time data. The software used to analyze and compare the researched methods is MATLAB/Simulink R2021b. The proposed method has performed better compared to both CSA and P&O for four conducted scenarios. Additionally, it was observed that the proposed method leads to less oscillation at the MPP point compared to P&O and CSA.

1. Introduction

In recent years, due to increasing environmental awareness and the limited nature of energy sources, interest in renewable energy sources has gradually increase [1]. Renewable energy sources provide many advantages such as low carbon emissions, sustainable resource use, and a cleaner environment [2]. Renewable energy obtained from many different sources such as solar energy, wind energy, hydroelectric energy, geothermal energy, and biomass energy is used in many areas including electricity generation, heating, cooling, lighting, and transportation [3]. As a result, renewable energy sources increase energy security, diversify energy supply, and provide economic benefits while reducing the use of traditional energy sources. Solar energy systems, being an important renewable energy source, have been increasingly preferred in recent years. Solar energy systems convert solar energy into electricity through solar panels. They are preferred due to their low maintenance costs, long lifespan, and eco-friendliness. In addition, solar energy systems increase energy independence, reduce energy costs, and ensure energy security. One of the most significant disadvantages of solar energy systems is the low efficiency of solar panels. Environmental factors such as irradiation and temperature changes cause fluctuations in the current and voltage values produced by the panel [4]. Maximum power point tracking (MPPT) algorithms are used to achieve maximum power (MP) from the PV panel in all environmental conditions [5]. MPPT is an effective solution to the disadvantages of solar energy systems. With MPPT technology, the efficiency of solar panels increases, and the systems can be used even in areas with less sunlight. There have been many different algorithms and methods proposed in the literature for MPPT studies. Some of these include perturb and observe (P&O) [6,7], incremental conductance algorithm (INC) [8], fuzzy logic (FL) [9], artificial neural networks (ANN) [10], particle swarm optimization (PSO) [11], cuckoo search algorithm.

Nomenclature	Abbreviation
PV	Photovoltaic
MPPT	Maximum power point tracking

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https://doi.org/10.1016/j.renene.2023.119470
Received 2 May 2023; Received in revised form 15 September 2023; Accepted 13 October 2023
Available online 14 October 2023
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(*continued*)

Nomenclature	Abbreviation
GPR	Gaussian process regression
STSMC	Super twisting sliding mode controller
CSA	Cuckoo search algorithm
P&O	Perturb and observe
MP	Maximum power
INC	Incremental conductance algorithm
FL	Fuzzy logic
ANN	Artificial neural networks
PSO	Particle swarm optimization
ACO	Ant colony optimization
PSC	Partial shading conditions
SRA	Search and rescue algorithm
HHO	Harris hawk optimization
SSA	Squirrel search algorithm
SFLA	Shuffled frog-leaping algorithm
SSA	Squirrel search algorithm
MWP	Most valuable player algorithm
WOA	Whale optimization algorithm
MCA	Musical chairs algorithm
SVM	Support vector machine
ECM	Equivalent circuit model
MSE	Mean Square Error
MAE	Mean Absolute Error
RMSE	Root-Mean-Square Error
R^2	Correlation of determination
SMC	Sliding mode control
SOSMC	Second order sliding mode control
GPR-STSMC	Gaussian process regression- Super twisting sliding mode controller

(CSA) [12], and ant colony optimization (ACO) [13]. The main aim of all these algorithms is to transfer the maximum power from the PV panel to the load. The P&O algorithm tracks the MPP point by measuring the instantaneous current and voltage values of PV panels. P&O is one of the most preferred algorithms among classical optimization algorithms due to its simple applicability and lack of complex mathematical calculations. However, in the case of partial shading conditions (PSC), P&O can stick to local points and therefore cannot perform MPPT. As a remedy, P&O is used together with alternative solution techniques in a hybrid manner under PSCs [14,15]. Alternatively, the INC algorithm tracks the maximum power point of the panels by using the current-voltage curves of the panels. Despite its advantages such as its fast response to environmental conditions, this algorithm cannot perform MPPT under PSCs. The poor performance of conventional optimization algorithms in PV systems operating under PSCs has made it necessary for researchers to use metaheuristic optimization algorithms. Meta-heuristic optimization is a method used to combine multiple heuristic optimization algorithms in the problem-solving process to achieve higher performance and a broader search space. Meta-heuristic optimization typically works by combining and comparing a set of candidate solutions to find the optimal solution. The most commonly used metaheuristic optimization algorithms for MPPT are PSO [16], CSA [17], ACO [18], search and rescue algorithm (SRA) [19], Harris hawk optimization (HHO) [20], BAT search [21], shuffled frog-leaping algorithm (SFLA) [22], squirrel search algorithm (SSA) [23], most valuable player algorithm (MWP) [24], whale optimization algorithm (WOA) [25], and musical chairs algorithm (MCA) [26]. Although metaheuristic optimization algorithms are quite successful in solving many optimization problems, their most significant disadvantages are their need for high computation power, long solution convergence time, and need for parameter tuning. Another MPPT method is FL and ANN-based artificial intelligence techniques [27,28]. In addition, hybrid methods using artificial intelligence-based algorithms together with classical or metaheuristic optimization algorithms have been presented in the literature for MPPT [29–31].

In this study, a new method (GPR-STSMC) has been proposed for MPPT both under non- PSCs and PSCs. In the first stage, several machine learning algorithms including Gaussian process regression (GPR) and Support vector machine (SVM) were trained to determine the most successful machine learning algorithm. With the selected machine learning algorithm, a model was created with temperature and irradiance values as inputs and voltage values as outputs. The dataset used in the model was created using hourly variations of temperature and irradiance values that occurred between April 1, 2021, and March 31, 2022, in the province of Erzurum, Turkey (https://power.larc.nasa.gov/data-access-viewer/). In the second stage, an error-based controller, the super-twisted sliding mode controller (STSMC), was used to optimize the duty cycle for transferring maximum power to the load. For all of these scenarios, a PV system was designed as shown in Fig. 1. The visual representation of the determined region is shown in Fig. 2.

This paper is structured as follows: Section 2 provides the modeling of photovoltaic panels and the I–V and P–V characteristics under partial shading conditions. Section 3 introduces SVR, GPR, hyperparameter optimization, and super twisting SMC. Section 4 presents the results and comparisons, and finally, Section 5 provides the conclusions.

2. Material-method

2.1. PV panel modeling and I–V, P–V characteristics under partial shading conditions

PV panels are composed of the combination of PV cells in series and parallel [32]. PV cells are semiconductor materials that convert solar energy into electrical energy. Different methods such as the equivalent circuit model (ECM), physical modeling, supervised learning, and artificial neural networks are used to model PV cells. PV panels, consisting of PV cells, are equipped with series, parallel, or series-parallel connections to achieve the desired power level. In addition, the power obtained from PV panels varies depending on environmental factors such as temperature and irradiance. Among these, ECM is the most commonly preferred due to its simple structure. The equivalent circuit of a single-diode ECM is shown in Fig. 3.

In Fig. 3, G represents the irradiance value and T represents the temperature value. The series resistance (R_S) represents the internal resistance of the cell and connection resistances. The parallel resistance (R_P) represents the leakage current resistance. The PV source element is used to model the power generation of the PV cell. The mathematical modeling of the PV cell is given by Equations (1)–(5) [33].

$$I_{pv} - I_d - I_p = I \tag{1}$$

$$I_d = I_0\left(\exp\left(\frac{V + IR_S}{nV_T}\right) - 1\right) \tag{2}$$

$$I_P = \frac{V + IR_S}{R_P} \tag{3}$$

$$I_{pv} = I_0\left(\exp\left(\frac{V + IR_S}{nV_T}\right) - 1\right) + \frac{V}{R_P} + I\left(\frac{R_S}{R_P} + 1\right) \tag{4}$$

$$I = I_{pv}\left(\frac{R_S}{R_P + R_S}\right) - I_0\left(\frac{R_S}{R_P + R_S}\right)\left(\exp\left(\frac{V + IR_S}{nV_T}\right) - 1\right) - \frac{V}{R_P + R_S} \tag{5}$$

where $''I_0''$ represents reserve saturation current (A), $''R_S''$ is series resistance, $''R_P''$ is parallel resistance and finally (n) represents the diode ideality constant. PV panels, consisting of PV cells, are equipped with series, parallel, or series-parallel connections to achieve the desired power level. In addition, the power obtained from PV panels varies depending on environmental factors such as temperature and irradiance.

High temperatures decrease the efficiency of the panel, while low temperatures increase its performance. This is because PV cells heat up more at high temperatures, increase the internal resistance, and decreasing the resistance at low temperatures. However, extremely low temperatures can damage the cells and even cause them to break. Fig. 4

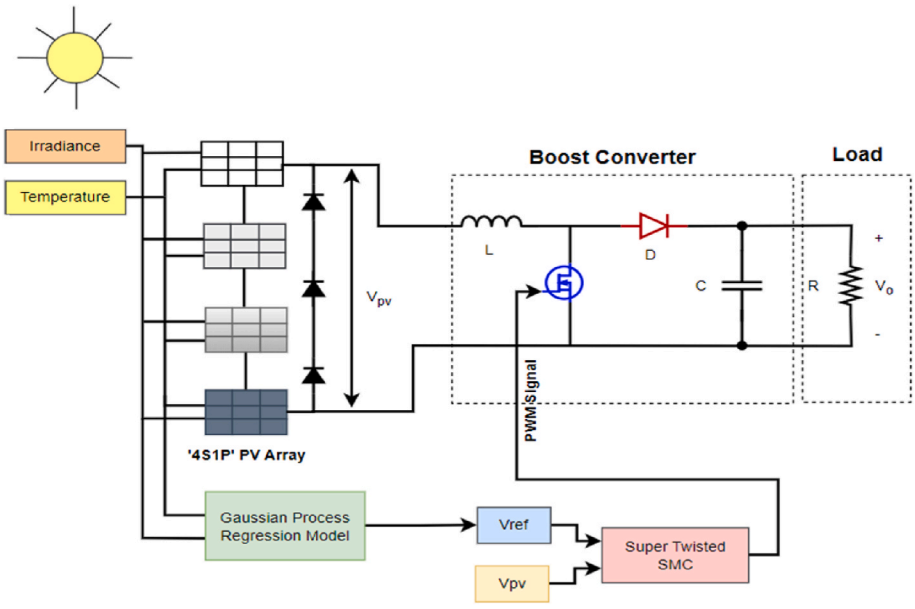


Fig. 1. Block diagram of the designed system.

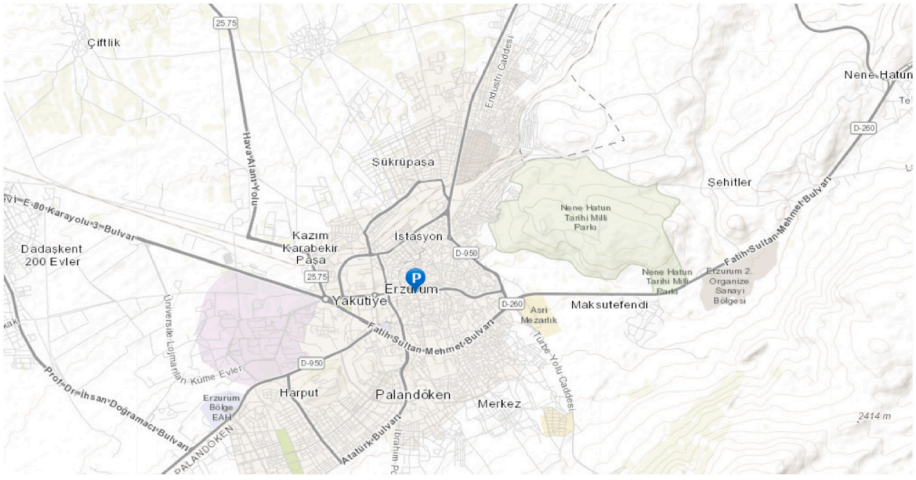


Fig. 2. Data region.

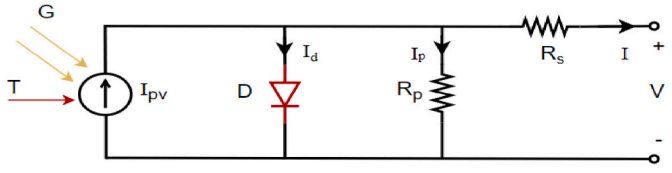


Fig. 3. Single diode PV cell model.

shows the effect of temperature changes on the I–V and P–V characteristics of a PV panel.

The performance of PV panels is directly related to the level of solar irradiance. As the solar irradiance increases, the electrical energy production of the PV panel also increases. This is because the solar rays hitting the surface of the panel release electrons and cause them to move. Therefore, the best performance of PV panels is achieved at high levels of solar irradiance. Fig. 5 shows the effect of irradiance variation on the I–V and P–V characteristics of the PV panel.

In this study, Aavid Solar ASMS-220P PV panel is used during experiments for which the characteristic are provided in Table 1 in the

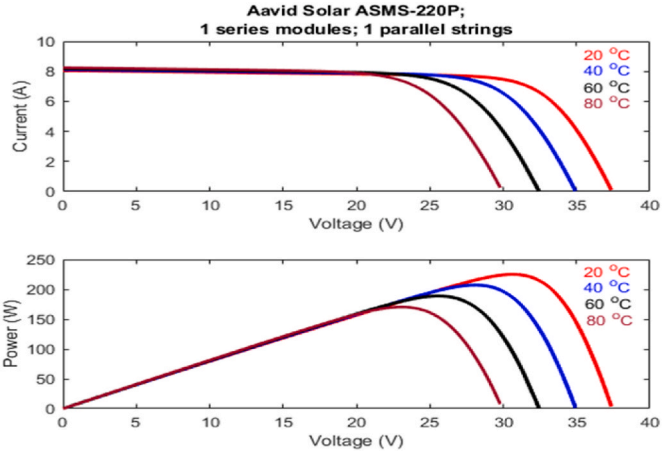


Fig. 4. P–V, I–V characteristics of PV cell at constant irradiation, variable temperature value.

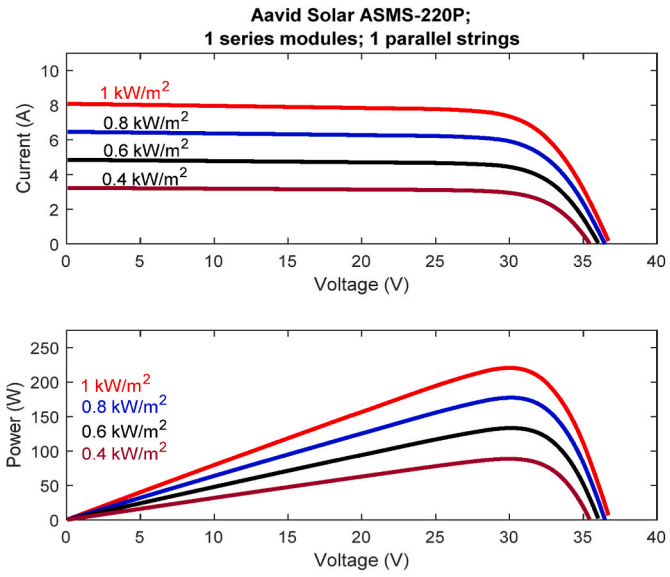


Fig. 5. P–V, I–V characteristics of PV cell at constant temperature, variable irradiance value.

Table 1
Parameters and their values of PV panel.

Parameters	Value
Maximum Power	220.5 W
Open Circuit Voltage	36.8 V
Voltage at MPP Vmp	30 V
Short-Circuit current Isc	8.08 A
Current at MPP Imp	7.35 A
Temperature coefficient of Isc (%/deg.C)	0.038465
Temperature coefficient of Voc (%/deg.C)	−0.3364
Cell per module (Ncell)	60

following.

2.2. Modelling of reference voltage signal process using machine learning algorithms

In this study, hourly temperature and irradiation data for Erzurum province for one year (01.04.2021–31.03.2022) are used to create models for the cases of partial shading and no shading. In the modeling process, support vector regression (SVR) and gaussian process regression (GPR) machine learning algorithms are used to predict the voltage value corresponding to the temperature and irradiation values. For this purpose, the data is first normalized using the following equation:

$$X_s = \frac{x-\mu}{\sigma} \tag{6}$$

where, x, μ and σ denote the input vector, average and standard deviation of input datasets.

2.2.1. Support vector regression (SVR) methodology

The SVR is an advanced machine learning technique for data categorization such as in engineering and financial time series forecasting domains and so on [34]. The purpose of this method is to seek a hyperplane to which the sum of the distance as from all data points is minimized. The training data of PV system is given as $x = [x_{irradiance}, x_{temperature}]$ while the output data is $y = [y_{Vref}]$. This regression model is as follows:

$$\psi(x) = \omega^T \bullet \varphi(x) + b \tag{7}$$

where $\psi(x)$ denote the predicted value by the SVR model and ω and b are constant parameters which are estimated by minimizing the regularized risk function in the following form.

$$\left.\begin{array}{c} \min\limits_{\omega,b,\xi_m,\xi_m^*} 0.5\|\omega\|^2 + C\sum\limits_{m=1}(\xi_m + \xi_m^*) \\ subject\ to\ y_m - \omega \bullet \varphi(x) - b \le \epsilon + \xi_m \\ w\varphi(x) + b - y_m \le \epsilon + \xi_m^* \\ \xi_m^* \ge 0 \end{array}\right\} \tag{8}$$

In this equation, C coefficient denote the regularization parameter. The ξ_m *and* ξ_m^* symbols are slack variables and represent the training error above ϵ and below $-\epsilon$, respectively. By minimizing the $0.5\|\omega\|^2$ regularization term, we prevent over-fitting and underfitting. In this way, the problem is converted into an optimization problem and the closed-form solution is obtained in terms of Lagrange multipliers the following explicit form:

$$\psi(x) = \sum_m (\alpha_m - \alpha_m^*)\kappa(x,\hat{x}) + b \tag{9}$$

where $\kappa(x,\hat{x})$ is the kernel function utilized in SVM which can take several forms such as which are listed in Table 2. One may extract from Table 2 that there are several parameters that need to be optimized to obtain best prediction performance.

2.2.2. Gaussian process regression (GPR) methodology

GPR is a powerful nonparametric, kernel-based machine learning technique that can predict nonlinear and complex relationships between the input and output variables of the process or system. GPR can make predictions by combining prior knowledge with a training dataset that contains input and output data. GPR is determined by the mean function $\mu(\boldsymbol{x})$ and the covariance functions $k(\boldsymbol{x},\boldsymbol{x}')$ as follows.

$$\left.\begin{array}{c} p(f|X) \sim N(\mu(x), \kappa(x,\hat{x})) \\ \mu(x) = \mathbb{E}[y(x)] \\ \kappa(x,\hat{x}) = \mathbb{E}[(y(x)-\mu(x))(y(\hat{x})-\mu(\hat{x}))] \end{array}\right\} \tag{10}$$

where $\mathbb{E}$ is the expectation function. Subsequently, the unknown function is defined with white noise,

$$y = f(x) + \varepsilon \tag{11}$$

where the mean function $\mu(\boldsymbol{x})$ is set to zero consequently, the output variable, which is affected noise, is given by $\varepsilon \sim N(0, \sigma_n^2)$. Moreover, joint probability distribution of training data can be written in matrix form as:

$$\left.\begin{bmatrix} y \\ y_p \end{bmatrix} \sim N\left(0, \begin{array}{cc} \boldsymbol{K}(\boldsymbol{x},\boldsymbol{x}) + \sigma_n^2\boldsymbol{I}_n & \boldsymbol{K}(\boldsymbol{x}^*,\boldsymbol{x})^T \\ \boldsymbol{K}(\boldsymbol{x}^*,\boldsymbol{x}) + \sigma_n^2\boldsymbol{I}_n & k(\boldsymbol{x}^*,\boldsymbol{x}^*) \end{array}\right)\right\} \tag{12}$$

where $\boldsymbol{K}(\boldsymbol{x},\boldsymbol{x}) = \boldsymbol{K}_n = (k_{ij})$ is a *nxn* positive definite covariance matrix and $\boldsymbol{I}_n$ is a *nxn* unit matrix. $\boldsymbol{K}(\boldsymbol{x}^*,\boldsymbol{x}) = [k(\boldsymbol{x}^*,\boldsymbol{x_1}), k(\boldsymbol{x}^*,\boldsymbol{x_2})\ldots k(\boldsymbol{x}^*,\boldsymbol{x_n}))]$ is the covariance matrix of the set $\boldsymbol{x}$ $(nx1)$ with new input test point $\boldsymbol{x}^*$.

The estimation of the next distribution is determined depending on the posterior distribution of GPR in the following form.

Table 2
SVR kernel functions.

Kernel Function	Formula
Linear	$\kappa(x,\hat{x}) = \langle x,\hat{x}\rangle$
Gaussian	$\kappa(x,\hat{x}) = e^{\gamma(x-\hat{x})^2}, \gamma > 0$
Polynomial	$\kappa(x,\hat{x}) = (\gamma\langle x+\hat{x}\rangle + c)^d, \gamma > 0$
Sigmoid	$\kappa(x,\hat{x}) = \tanh(\gamma\langle x+\hat{x}\rangle + c), \gamma > 0$

$$p(y_p|\boldsymbol{x}^*,\boldsymbol{x},y) = N\Big(\boldsymbol{K}(\boldsymbol{x}^*,\boldsymbol{x})\left[\boldsymbol{K}(\boldsymbol{x},\boldsymbol{x}) + \sigma_n^2\boldsymbol{I}_n\right]^{-1}\boldsymbol{y}, \boldsymbol{K}(\boldsymbol{x}^*,\boldsymbol{x}^*) - \boldsymbol{K}(\boldsymbol{x}^*,\boldsymbol{x})\left[\boldsymbol{K}(\boldsymbol{x},\boldsymbol{x}) + \sigma_n^2\boldsymbol{I}_n\right]^{-1}\boldsymbol{K}(\boldsymbol{x}^*,\boldsymbol{x})^T \tag{13}$$

Equation (13) is calculated according to the Bayesian law and the covariance function is called as kernel function and can be expressed in different form as shown in Table 3.

In the GPR approach, complex input and output relationships can be modelled with limited number of hyperparameters. Therefore, the hyperparameters of both SVR and GPR models were optimized using the Bayesian optimization approach.

2.3. Hyperparameter optimization

The precise computation of hyperparameters is critical for the accurate prediction performance of machine learning algorithms. The determination of optimal hyperparameters significantly improves the prediction performance of the designed model. Therefore, the hyperparameters must be finetuned correctly to achieve optimal predictive ability. However, finding the precise values of optimal parameters is a complex and difficult task due to the need for tedious investigation over the parameter space.

Bayesian optimization is an evolved effective parameter search algorithm for configuring hyperparameters and uses a surrogate model such as the Gaussian process, random forest and an acquisition function unlike the grid search method [35]. In this study, the Bayesian optimization method is used to find the global minimum of an unknown function $f(x)$:

$$x^* = argminf(x), x \in A \tag{14}$$

where A is the hyperparameter search space of x, and $f(x)$ is defined as a cost function that must be optimized. We first utilize, the Bayesian optimization method is utilized to optimize the selection of kernel functions, C coefficient and γ of the SVR model and then the kernel function and value of σ_f are optimized for the GPR.

2.4. Evaluation of developed machine learning model performance

Mean square error, mean absolute error, root mean square error, and correlation of determination are provided in Table 4 as prediction performance criteria of the developed models.

Table 3
SVR kernel functions for GPR model.

Kernel Function	Mathematical Expression
Exponential Function	$k_{Exp}(\boldsymbol{x},\boldsymbol{x}') = \sigma_f^2 e^{\left(-\frac{\sqrt{(x-x')^T(x-x')}}{\sigma_l}\right)}$
Matern Function (5/2)	$k_m(\boldsymbol{x},\boldsymbol{x}') = \sigma_f^2\left(1 + \frac{\sqrt{5}\sqrt{(x-x')^T(x-x')}}{\sigma_l} + \frac{5(\sqrt{(x-x')^T(x-x')})^2}{3\sigma_l^2}\right)e^{\left(-\frac{\sqrt{5}\sqrt{(x-x')^T(x-x')}}{\sigma_l}\right)}$
Matern Function (3/2)	$k_m(\boldsymbol{x},\boldsymbol{x}') = \sigma_f^2\left(1 + \frac{\sqrt{3}\sqrt{(x-x')^T(x-x')}}{\sigma_l}\right)e^{\left(-\frac{\sqrt{3}\sqrt{(x-x')^T(x-x')}}{\sigma_l}\right)}$
Squared exponential Function	$k_S(\boldsymbol{x},\boldsymbol{x}') = \sigma_f^2 e^{\left(-\frac{1}{2}\frac{(x-x')^T(x-x')}{\sigma_l^2}\right)}$

Table 4
Prediction performance criteria of the developed models.

Performance Evaluation Criteria	Mathematical Expression
Mean Square Error (MSE)	$MSE\ (V_{ref\ actual}, \widehat{V}_{ref\ actual}) = \frac{\sum_{k=1}^{N}(V_{ref\ actual} - \widehat{V}_{ref\ actual})}{N}$
Mean Absolute Error (MAE)	$MAE\ (V_{ref\ actual}, \widehat{V}_{ref\ actual}) = \frac{\sum_{k=1}^{N}\lvert V_{ref\ actual} - \widehat{V}_{ref\ actual}\rvert}{N}$
Root-Mean-Square Error (RMSE)	$RMSE(V_{ref\ actual}, \widehat{V}_{ref\ actual}) = \sqrt{\frac{1}{N}\sum_{k=1}^{N}(V_{ref\ actual} - \widehat{V}_{ref\ actual})^2}$
Correlation of determination (R^2)	$R^2\ (V_{ref\ actual}, \widehat{V}_{ref\ actual}) = 1 - \left(\frac{\sum_{k=1}^{N}(V_{ref\ actual} - \widehat{V}_{ref\ actual})^2}{\sum_{k=1}^{N}(V_{ref\ actual} - \widehat{V}_{ref\ actual})^2}\right)$

Fig. 6 shows the reference voltage values and the predicted voltage values obtained using SVR and GPR methods during the model building process. The results of SVR and GPR according to performance criteria such as $MSE, RMSE, MAE$ and R^2 are shown in Fig. 7. $V_{ref\ actual}$ and $\widehat{V}_{ref\ actual}$ represent the actual (desired) value, predicted (fitted) value by the developed models. One can see that, in terms of all performance criteria, GPR outperformed SVR and thus, GPR was preferred for determining the voltage value in the study. Table 5 shows the results of these two machine learning methods.

2.5. Super twisting sliding mode control (STSMC)

The super twisting controller, a second order sliding mode control (SOSMC) algorithm is successfully implemented in the determination of the optimum duty cycle to transfer maximum power to the load, which keeps the reference value of the PV voltage at the desired level in the absence and presence of partial shading.

It is well known that the main problem in the implementation of

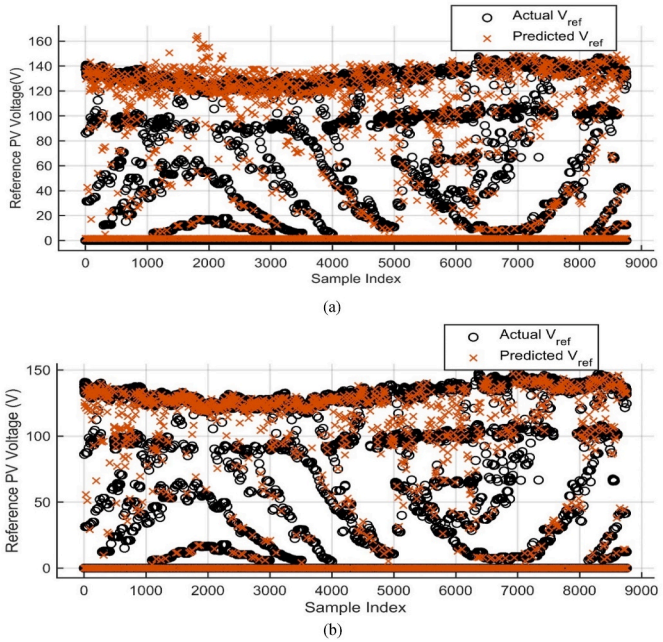


Fig. 6. Comparison between actual and predicted reference PV voltage for (a) SVR method (b) GPR method.

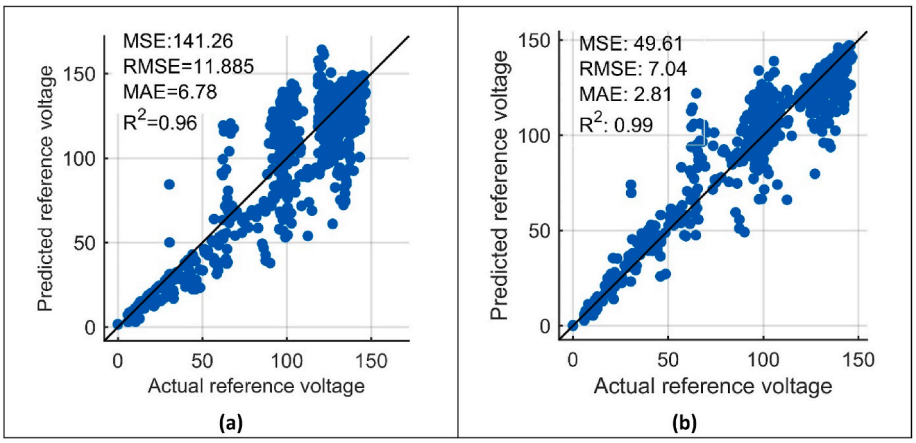


Fig. 7. Comparison between the actual and predicted reference voltage by the (a) SVR and (b)GPR models.

Table 5
Comparison of SVR and GPR methods.

Performance Criteria	MSE	RMSE	MAE	R^2
SVR	141.26	11.885	6.78	0.96
GPR	49.61	7.04	2.81	0.99

sliding mode control (SMC) is the chattering phenomenon caused by the use of discontinuous switching control functions. Therefore, the Super Twisting SMC algorithm can be used instead of traditional SMC particularly for the first order systems. The super twisting sliding mode control is one of the most commonly used high orders sliding mode control strategies and control term of this algorithm is composed of two terms: the first term is the integral of a discontinuous control action and second term is a continuous function of s. The control law is written as follows:

$$\left.\begin{aligned} u(t) &= u_1(t) + u_2(t) \\ \dot{u}_1(t) &= -\gamma \mathrm{sign(s)} \\ u_2(t) &= -\varphi |s|^{\frac{1}{2}} \mathrm{sign(s)} \end{aligned}\right\} \tag{15}$$

where γ and φ are controller design parameters calculated from the corresponding sufficient conditions for finite time convergence of the sliding manifold in the following form:

$$\left.\begin{aligned} W &> \frac{\rho}{\tau_m} \\ \varphi &\geq \sqrt{\frac{4\rho}{\tau_m}\frac{\tau_m(W+\rho)}{\tau_m(W-\rho)}} \end{aligned}\right\} \tag{16}$$

where ρ and τ_m are constant parameters and W and φ denote controller gains.

3. Simulation results

A system consisting of a PV panel, a DC-DC converter, an MPPT controller, and a load was designed to evaluate the performance of the proposed method. PV panels typically generate a direct current (DC) voltage level that is lower than the voltage level required by the load. Therefore, a boost-type converter (such as a DC-DC booster) is used to increase the voltage level produced by the PV panel and transfer the increased voltage to the load after adjusting it to the desired levels. This converter prepares the electrical energy produced by the PV panel for use. MPPT controllers are used to ensure that the system operates at the highest possible efficiency under varying irradiance or temperature conditions. MPPT controllers control the duty cycle of semiconductor switches (Mosfet, IGBT) to transfer maximum power to the load. The flowchart of the designed system is shown in Fig. 8. The circuit parameters of the boost-type converter are given in Table 6.

The performance of the GPR-STSMC was evaluated by considering three different scenarios based on high (600–900W), medium (300–600W), and low (0–300W) power levels, in the absence or presence of PSCs. The first scenario was carried out considering the case of no PSCs and working at a high-power level. The second scenario was applied for the case where PSCs are presented and a medium power level is implemented. The third scenario was created using a low power level again with PSCs. The P–V characteristics of the PV panels for all scenarios are shown in Fig. 9.

3.1. First scenario

In order to compare the performance of the GPR-STSMC in the first scenario, a PV system with a '4S1P' connection structure was used. The irradiance values of the system are respectively, set to 875.85 W/m^2, 870.65 W/m^2, 899.98 W/m^2, and 904.16 W/m^2, while the temperature values are respectively, set to 23.52°C, 25.01°C, 27.01°C, and 22.97°C, using real-time data. In addition, the first scenario was performed for the high level MPP (786.57W) in the non-PSCs. Results such as load voltage, load current, power dissipated in the load, PV voltage, and PV current for the GPR-STSMC, P&O and CSA are evaluated separately. Fig. 10 shows the power consumption values on the load for the GPR-STSMC, P&O, and CSA algorithms. GPR-STSMC reached the MPP point in a shorter convergence time compared to both classical and metaheuristic optimization algorithms in the case where no PSCs are used. The power consumption values on the load for this case are given in Table 7.

Fig. 11 a shows the variation of the load voltage over time for the application of GPR-STSMC, P&O and CSA algorithms. The voltage values for GPR-STSMC, P&O and CSA are listed in Table 7. It can be observed that there are fluctuations in the voltage values for the CSA algorithm due to its search structure. As the CSA algorithm converges to the MPP point, the voltage fluctuation decreases. Fig. 11 b shows the variation of the current passing through the resistive load over time for the GPR-STSMC, P&O and CSA. The algorithms converge to approximately the same current value at different times. The current values for this scenario are also given in Table 7. In Fig. 11 c, the voltage values generated by the PV panel for GPR-STSMC, P&O and CSA are shown for the first scenario. The P&O algorithm causes oscillation at the MPP point because it changes the duty cycle by ″Δd″ amounts. On the other hand, the convergence approach used to obtain the optimal duty cycle in the CSA algorithm causes voltage fluctuations. In the proposed model, the reference voltage value is generated with GPR and controlled with STSMC, resulting the desired voltage value after minimum oscillations in a shorter convergence time compared to other algorithms. The PV voltage values for this scenario are summarized in Table 7. Fig. 11

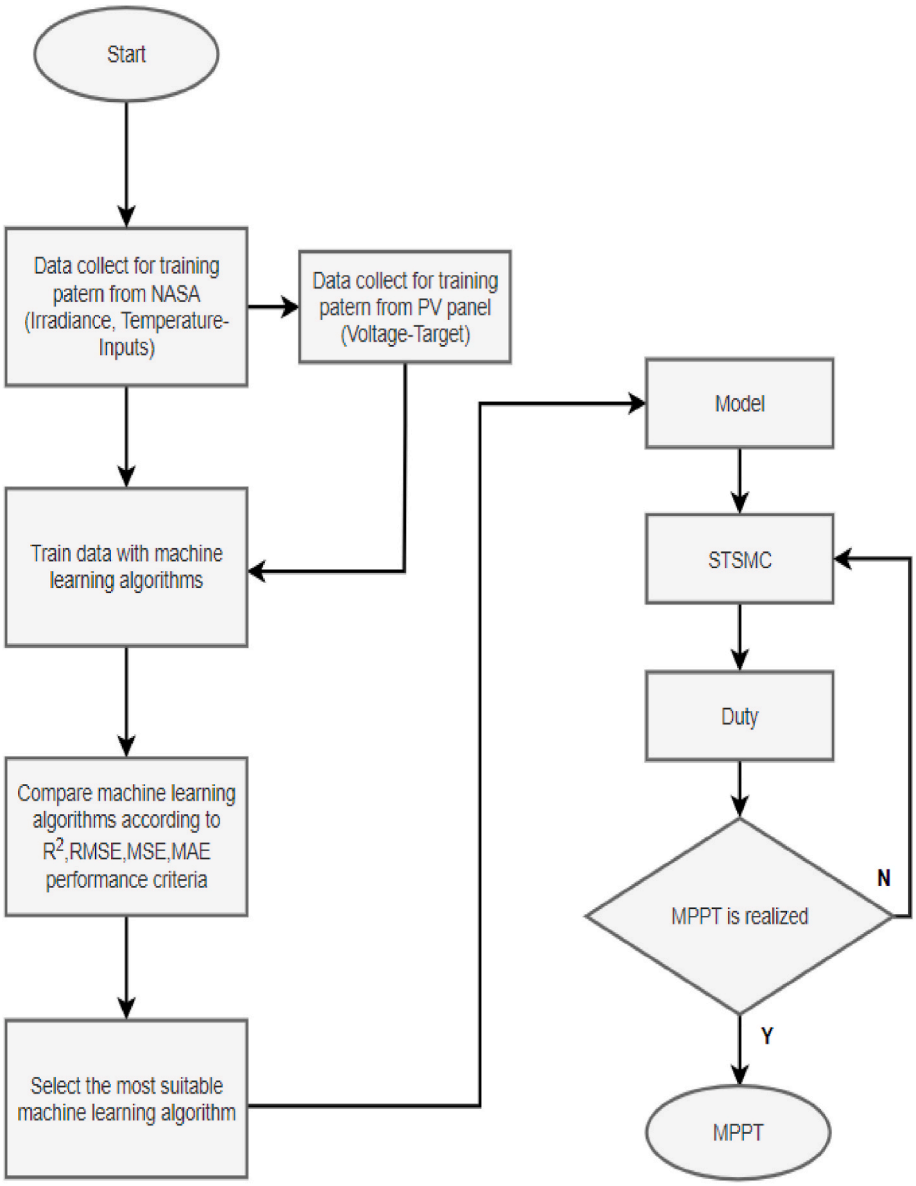


Fig. 8. The flowchart of the proposed system.

Table 6
The parameters of boost converter.

Parameters	Value
Inductance (L)	2.9 *mH*
Capacitors (C_{in} and C)	1 μF- 1.56 mF
Load	50 Ω
Switching Frequency	50 kHz

d shows the current values generated by the PV panels. As with the voltage values, the GPR-STSMC reaches the desired current value more stably compared to CSA and P&O. The current values generated by the PV panels for the GPR-STSMC, P&O and CSA are presented in Table 7.

3.2. Second scenario

In the second scenario, unlike the first scenario, a negative temperature value was observed. The effect of temperatures below zero in the region where the data was collected has contributed to the negative selection of temperature values. The irradiation values for the panels are

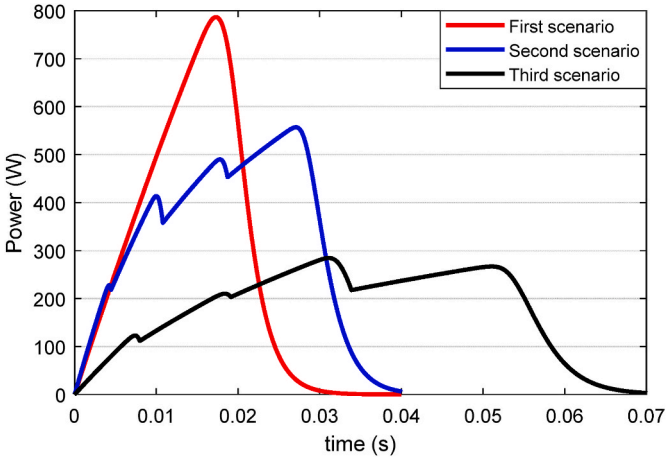


Fig. 9. P–V characteristics of PV panels for all scenarios.

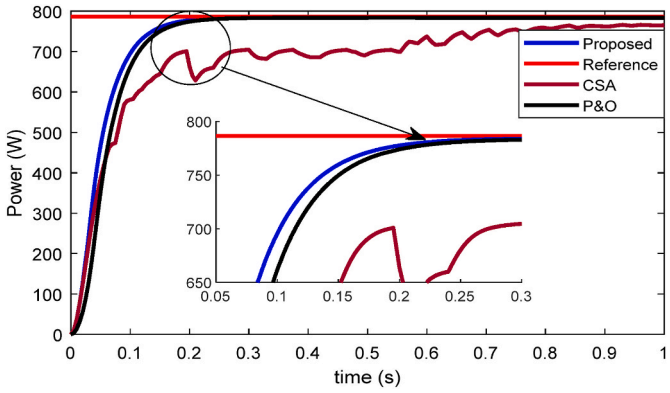


Fig. 10. Load power.

determined as 610.80 W/m^2, 511.27 W/m^2, 976.61 W/m^2, and 810.70 W/m^2 and the temperature values are $-1.13°C$, $-3.14\,°C$, $-6.51\,°C$, and $-7.35°C$, respectively. GPR-STSMC, P&O and CSA are evaluated in terms of performance criteria for this situation as well. The variation of power transferred to the load under PSCs over time is shown in Fig. 12 for the second scenario. It can be observed that the metaheuristic optimization algorithm CSA was not able to reach the MPP exactly within the specified time frame, while the classical optimization algorithm P&O was trapped at a local maximum point. GPR-STSMC, on the other hand, achieved MPPT in a shorter time than the other algorithms under PSCs.

The current and voltage values generated by PV panels and the

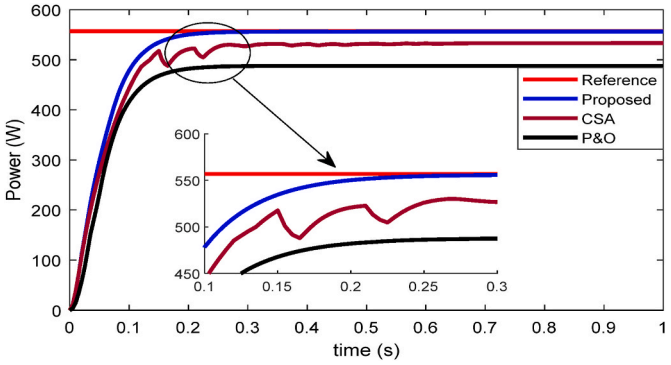


Fig. 12. Load power.

Table 7
Simulation results of algorithms for different PSCs using real-time data.

Scenarios	Algorithm	GMPP(W)	Measured Power (W)	Convergence Time (s)	Load Voltage (V) Load Current (A)	Efficiency (%)
First scenario	GPR-STSMC	786.57	784 .5822	0.17	198.066 3.9612	**99.747**
	CSA	786.57	784.54	1.21	198.04 3.9614	99.741
	P&O	786.57	783.84	0.19	197.97 3.9593	99.652
Second scenario	GPR-STSMC	556.9	556.24	0.18	166.77 3.3353	**99.881**
	CSA	556.9	534.2134	1	163.43 3.2687	95.92
	P&O	556.9	487.8464	–	156.18 3.123	87.60
Third scenario	GPR-STSMC	284.4	284.0486	0.19	119.17 2.3834	**99.876**
	CSA	284.4	280.702	0.325	118.489 2.369	98.699
	P&O	284.4	283.2951	0.245	119.016 2.3803	99.611

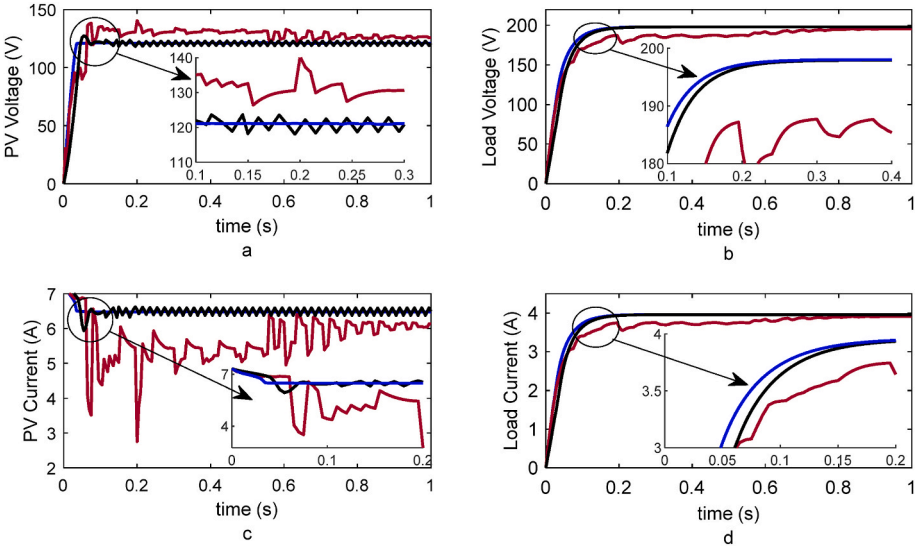


Fig. 11. a) PV voltage b) Load voltage c) PV current d) Load current.

current and voltage values on the load for the second scenario are shown in Fig. 13. Compared to the CSA and P&O algorithms, it has been observed that GPR-STSMC has a shorter settling time and much lower level of oscillations. Optimizing the duty cycle in CSA caused momentary changes in voltage and current values. Moreover, changing the duty cycle by the "Δd" ratio caused oscillation in the P&O algorithm results.

3.3. Third scenario

In the third scenario, unlike the other scenarios, a situation where both irradiance and temperature values are low has been observed. In this case, the irradiance values are determined as 245.81 W/m^2, 364.60W/m^2, 418.00 W/m^2, and 539.80 W/m^2, and the temperature values are determined as $-5.75°C$, $-4.10°C$, $-2.31°C$, and $1.03°C$. The power values over time for GPR-STSMC, P&O, and CSA algorithms applied in the third scenario are shown in Fig. 14. The disadvantage of the P&O algorithm (it was trapped at the local maximum point under PSCs) is eliminated by setting the reference voltage value as the starting voltage level. The difference between this scenario and the second scenario is that the starting value of the reference voltage for the P&O algorithm is a given parameter. Considering all these conditions, the proposed GPR-STSMC achieved a shorter convergence time compared to the CSA and P&O in this scenario.

The results of the GPR-STSMC and the CSA, P&O algorithms are shown in Fig. 15. Thanks to the GPR-based reference voltage generation in the first stage of the proposed model, the STSMC algorithm achieves MPPT much faster than the CSA and P&O. The improved P&O algorithm also reaches the global maximum value by surpassing the local maximum point. The results of the GPR-STSMC and the CSA and P&O are given in Table 7.

The comparison of convergence times for GPR-STSMC, P&O and CSA in different scenarios is illustrated in Fig. 16, while the comparison of their efficiencies is shown in Fig. 17.

3.4. Fourth scenario

The scenarios we realized were determined by using real-time data (hourly data), considering the situations without partial shading and situations with PSCs. The fact that the temperature change is not considered in the literature and the climatic conditions of the Erzurum region have been effective in determining the scenarios. Since there is no production in the early hours of the day and night when the irradiation values are zero, those time zones were not considered in the simulation.

In order to test the transient response of the proposed model and to guarantee that the model will behave appropriately, hourly data from a random day from "https://power.larc.nasa.gov/data-access-viewer/"

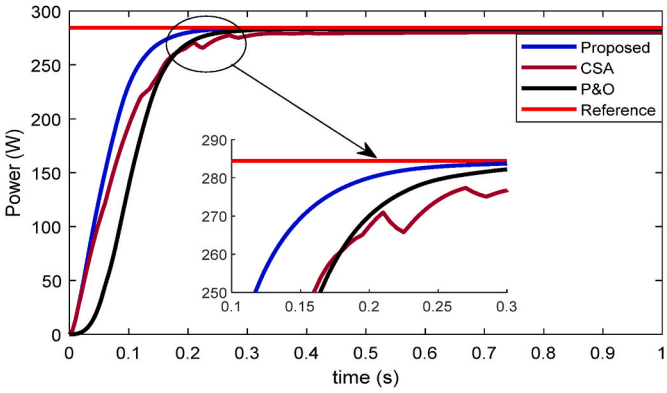


Fig. 14. Load power.

was used in fourth scenario. Additionally, in the fourth scenario, temperature and radiation changes during the day were examined by scaling each hourly time period to 1 s time period.

In Fig. 18, the variation of the irradiation values of the four panels with respect to time is shown.

The variation of temperature values with respect to time is given in Fig. 19.

The estimated voltage values produced by the model in Fig. 20 over time are shown in Fig. 21, respectively, the power consumed on the load.

The results of the fourth scenario are given in Table 8 for proposed algorithm.

Considering the dynamically changing environmental conditions, the proposed method performed MPPT quickly and efficiently for all time periods.

The evaluation of the GPR-STSMC compared to other studies that performed MPPT monitoring is presented in Table 9.

4. Conclusions

PSCs reduce the efficiency of solar panels. By-pass diodes are used to prevent PSCs, which avoid the shaded panels and allow the system to continue operating. However, by-pass diodes cause multiple local maxima to occur in the P–V voltage characteristic of PV systems. Classic optimization algorithms cannot perform MPPT since they are trapped at one of the local maxima. Metaheuristic optimization algorithms, on the other hand, can perform MPPT in PSCs. The disadvantage of these algorithms is their long convergence time due to the complex mathematical calculations involved. In this study, a new MPPT method has been proposed that achieves shorter settling times and significantly

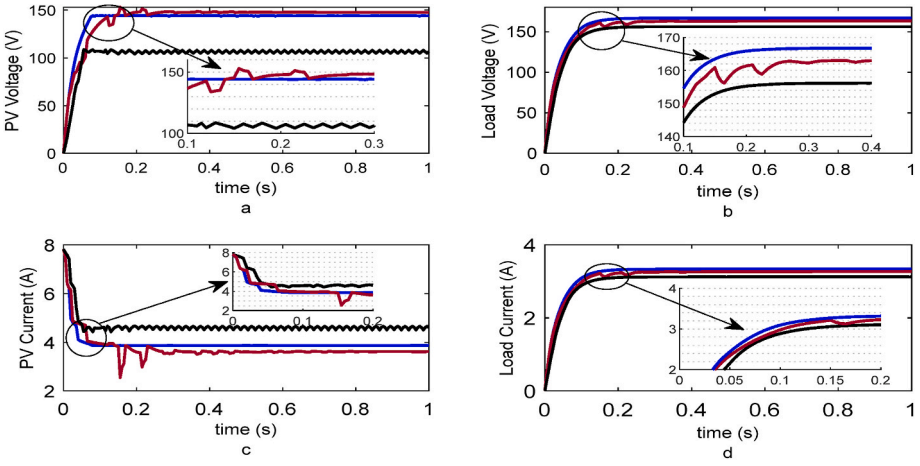


Fig. 13. a) PV voltage b) Load voltage c) PV current d) Load current.

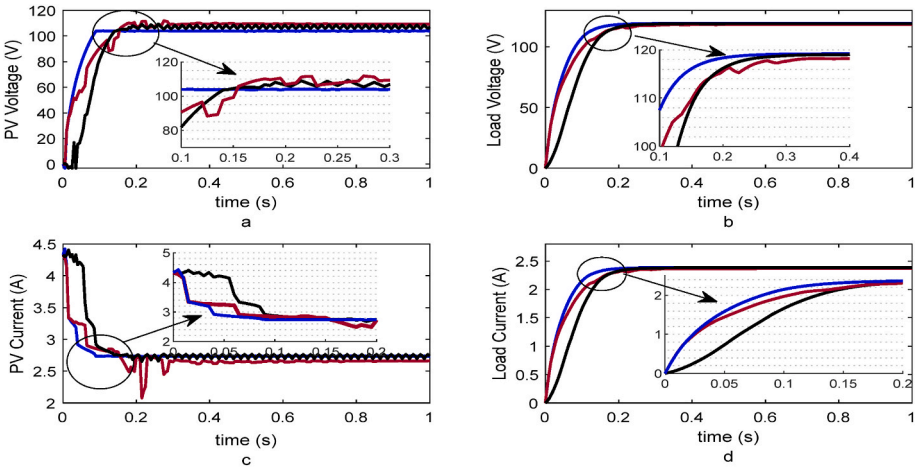


Fig. 15. a) PV voltage b) Load voltage c) PV current d) Load current.

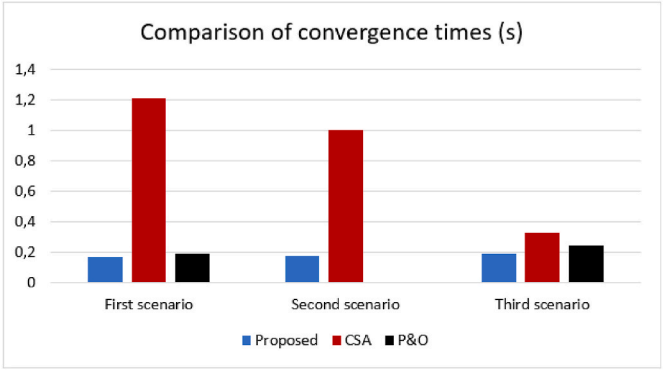


Fig. 16. Comparison of convergence times.

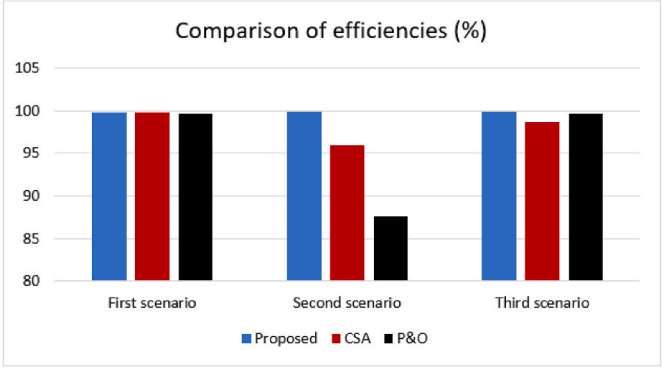


Fig. 17. Comparison of efficiencies.

reduces oscillations both under PSCs and non- PSCs. The proposed method is formed by the combination of GPR and STSMC. The training process of the real-time one-year data collected from the region (Erzurum/Turkey) determined by machine learning algorithms was carried out in the first stage. It was observed that GPR provided more successful results compared to other machine learning algorithms. In the second stage, MPPT operation was performed using the STSMC controller with the reference voltage value and the instantaneous PV panel output voltage obtained through the model created in the first stage. To evaluate the performance of the GPR-STSMC, a classic and a metaheuristic optimization algorithm were selected. It was observed that the GPR-STSMC reached the MPP point in a shorter time compared to both

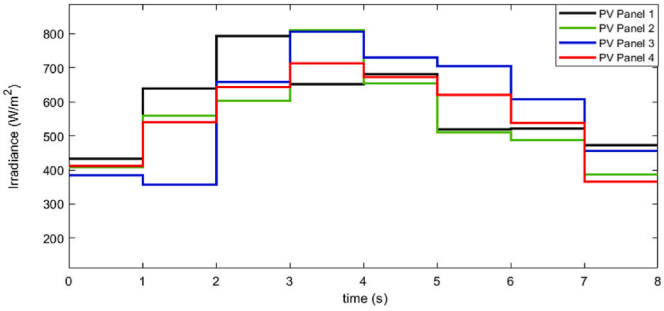


Fig. 18. PV panel irradiation values scaled so that a 1-h time period corresponds to a 1-s time period.

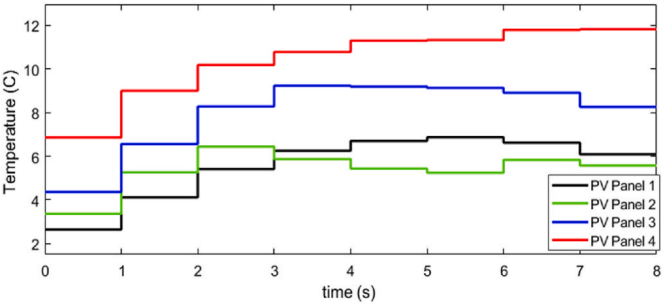


Fig. 19. PV panels temperature values scaled so that a 1-h time period corresponds to a 1-s time period.

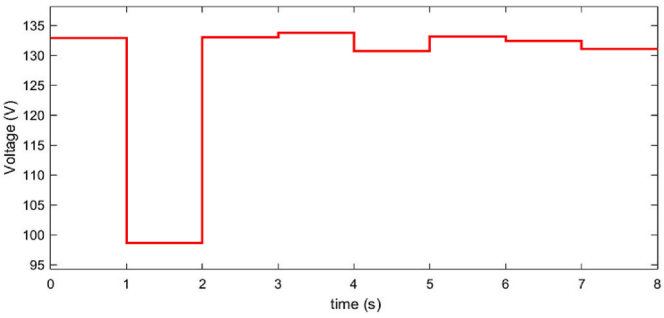


Fig. 20. Reference voltage values produced by the model.

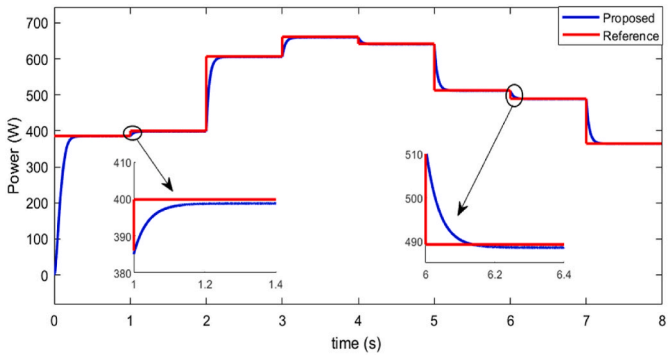


Fig. 21. Load power.

Table 8
Simulation results of GPR-STSMC.

Time (s)	GMPP (W)	Measured Power (W)	Convergence Time (s)	Efficiency (%)
0–1	386.04	385.62	0.221	99.891
1–2	399.85	398.76	0.026	99.727
2–3	606.76	605.6	0.114	99.808
3–4	661.15	659.75	0.059	99.788
4–5	641.85	640.57	0.012	99.800
5–6	512.40	511.64	0.095	99.851
6–7	489.25	488.55	0.031	99.856
7–8	364.70	364.34	0.108	99.901

Table 9
Performance comparison of the proposed method with algorithms that track MPPT.

Criteria	Convergence Speed	Input Var.	Conditions	Efficiency
Method in P&O [36]	Low	Voltage Current	Variable Atmospheric Conditions	Low
Method in INC [37]	Fast	Voltage Current	Variable Atmospheric Conditions	Low
Method in FLC [38]	Fast	Irradiance Temperature	Variable Atmospheric Conditions	High
Method in NN [39]	Fast	Irradiance Temperature	Variable Atmospheric Conditions	High
Method in CSA [17]	Fast	Voltage Current	Partial Shading	High
Method in PSO [11]	Medium	Voltage Current	Partial Shading	High
Method in super twisting SMC [40]	Fast	Voltage Current	Variable Atmospheric Conditions	High
Method in new P&O [14]	Fast	Voltage Current	Partial Shading	High
Method in adaptive FLC [28]	Fast	Irradiance Temperature	Variable Atmospheric Conditions	High
Method in ANFIS [41]	Fast	Voltage Current	Variable Atmospheric Conditions	Medium
Method in Fractional Open Circuit Voltage [42]	Medium	Irradiance Temperature	Variable Atmospheric Conditions	Medium
GPR-STSMC	Fast	Irradiance Temperature	Partial Shading	High

P&O and CSA algorithms for the three scenarios conducted. In addition, it has been observed that the proposed method reacts quickly to dynamically changing environmental conditions and performs MPPT with high efficiency.

CRediT authorship contribution statement

Mehmet Yılmaz: Conceptualization, Methodology, Software, Writing- Original draft ; Alirıza KALELİ Data curation, preparation. Software, Validation, Supervision ; M. Fatih Çorapsız: Conceptualization, Methodology, Software, Supervision, Reviewing and Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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